Natural Disaster Severity Prediction via Leakage-Aware Boosted Trees and Temporal Feature Engineering

Data Mining Final Project, 2026 Spring

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***Abstract*—Group ID: 5. Members: Hsin-Yu Chen (314551141, sychen.cs14@nycu.edu.tw), Wei-Hsin Hung (314553021, velda.cs14@nycu.edu.tw), and Sky Shih-Kai Hong (314554024, sky.cs14@nycu.edu.tw). Our code is available at GitHub: https://github.com/skyhong2002/disaster-severity-prediction.**

This project predicts weekly drought severity scores for 2,248 regions from historical meteorological observations. The central difficulty is not only multi-horizon forecasting, but also preventing target leakage from weekly score labels while adapting to train-test season and weather shift. We implement direct horizon models using LightGBM, XGBoost, and CatBoost, with calendar, lag, rolling, exponentially weighted, drought-proxy, climatology, and 91-day-gapped score-history features. The best reproducible reportable submission is a 35/35/30 LightGBM/XGBoost/CatBoost ensemble with Kaggle public MAE 0.8124. A separate already-submitted public-chase horizon hedge reaches public MAE 0.7915 and crosses Baseline 3, but we keep it separate from the method claim because it was selected by public leaderboard feedback rather than by a new training algorithm.

I. Project Summary (max 1 page)

A. Goal

The competition task is to predict the drought severity score for each region over the next five consecutive weeks after the 91-day test observation window. Scores are on a 0–5 scale, with larger values indicating more severe drought. The submission must contain one row per `region_id` and five numeric columns, `pred_week1` through `pred_week5`. The evaluation metric is mean absolute error (MAE).

B. Approach

We frame the task as supervised direct multi-step regression. For each horizon $h \in \{1, \dots, 5\}$, the training label is the real weekly score h weeks after a forecast origin. The final pipeline builds a time-safe feature table from raw weather, recent and long-window weather summaries, calendar variables, region-month climatology anomalies, and score-history features shifted at least 91 days. The final reportable model is a convex ensemble of three boosted-tree families: LightGBM, XGBoost, and CatBoost. We chose this design because tabular gradient boosting remains highly competitive on heterogeneous structured features, while being easier to audit than large neural sequence models.

C. Related Work and SOTA Context

Our method is closest to the competition-style tabular forecasting tradition: XGBoost [1], LightGBM [2], and CatBoost [3] are strong baselines for mixed continuous/categorical data and sparse feature interactions. Multi-horizon time series research has also advanced rapidly through interpretable neural forecasters such as TFT [4], basis/interpolation models such as N-BEATS [5] and N-HiTS [6], efficient Transformer variants such as Informer [7], Autoformer [8], FEDformer [9], PatchTST [10], TimesNet [11], and iTransformer [12], as well as simpler alternatives such as DLinear [13] and TSMixer [14]. Recent foundation models, including TimesFM [15], Chronos [16], Moirai [17], and later MoE-scale systems such as Time-MoE [18] and Timer-S1 [19], show the direction of SOTA zero-shot forecasting, but our dataset has a specific supervised target, high-cardinality regions, and strict leakage constraints. In drought forecasting, prior reviews emphasize that meteorological drought is nonlinear, nonstationary, and often benefits from hybrid or ensemble machine learning models [20]. The U.S. Drought Monitor literature also motivates our ordinal severity setting: drought categories combine multiple physical indicators and expert judgment, and transitions are usually incremental but can change rapidly under heat and rainfall shocks [21].

II. Proposed Method (1-3 pages)

A. Problem Formulation

Let $\mathcal{R}$ be the set of regions and let $\mathbf{x}_{r,t} \in \mathbb{R}^D$ be the meteorological feature vector for region r on day t. Weekly drought labels are observed only on score days, so most daily rows have missing labels. Given the final test window $\mathcal{T}_r^{test}$ of 91 days for region r, we predict five future weekly scores:

$$\hat{\mathbf{y}}_r = (\hat{y}_{r,1}, \hat{y}_{r,2}, \hat{y}_{r,3}, \hat{y}_{r,4}, \hat{y}_{r,5}). \tag{1}$$

The target metric is

$$\text{MAE} = \frac{1}{5|\mathcal{R}|} \sum_{r \in \mathcal{R}} \sum_{h=1}^{5} |\hat{y}_{r,h} - y_{r,h}| . \tag{2}$$

B. Leakage-Aware Feature Construction

The most important engineering constraint is that score labels inside the 91-day blind window are unavailable at inference time. Therefore all features that depend on `score` are computed only from history ending at least 91 days before the forecast origin. This rule also applies to validation: blind backtests explicitly mask the 91-day window before rebuilding features. Within this contract, the feature table contains:

- **Weather lags**: shifted daily values at 7, 14, 21, 28, 35, 42, and 49 days.
- **Rolling/EWM weather**: rolling mean and standard deviation over 7–35 days and exponentially weighted means with spans 3, 7, and 14.
- **Long drought proxies**: 30/90/180-day precipitation sums, dry-day counts, dry-day share, max dry streak, temperature means, dew-point depression, wet-bulb depression, and related heat/dryness indices.
- **Calendar and seasonality**: month, quarter, day-of-year, week-of-year, and cyclic sine/cosine encodings.
- **Climatology anomalies**: region-month normal weather and standardized deviations from those local seasonal norms.
- **Blind-gap score history**: lagged score summaries shifted by at least 91 days, so the model sees long-term severity context but not hidden test-window labels.

Several early experiments used names containing `two_stage`; these names are retained only for artifact compatibility. The active implementation is a direct-horizon supervised pipeline, not a separate score-reconstruction system.

C. Model Families

For each horizon, we train one independent regression model per family rather than an autoregressive five-week sequence model. For a feature vector $\mathbf{z}_{r,t}$ at forecast origin t, each model family learns an additive tree function

$$F_h(\mathbf{z}_{r,t}) = F_{0,h} + \sum_{m=1}^{M} \eta_m f_{m,h}(\mathbf{z}_{r,t}), \tag{3}$$

where each $f_{m,h}$ is a regression tree fitted to the residual gradient of the current ensemble. The three libraries share this boosting view, but their tree structure and regularization are different.

LightGBM. LightGBM is our primary tabular anchor. It first quantizes continuous inputs into histogram bins, then searches split gains on those bins instead of on all raw values [2]. Its trees are grown leaf-wise: after every split, the algorithm expands the current leaf with the largest estimated loss reduction. This best-first structure can form deep, specialized rules such as "very low 90-day precipitation in a specific seasonal context," which is useful for our many lag, rolling, drought-proxy, and climatology features. The risk is overfitting, so the implementation relies on constraints such as learning rate, number of leaves, minimum leaf data, and feature subsampling.

XGBoost. XGBoost also builds an additive ensemble of CART-style regression trees, but the split search is driven by a second-order regularized objective [1]. At each boosting step it approximates the loss with gradients and Hessians, penalizes tree complexity through the number of leaves and leaf weights, and then applies shrinkage and row/column subsampling. In practice this gives a more conservative, depth-controlled tree structure than LightGBM. It often captures similar nonlinear interactions, but with different split choices, making it a strong diversity partner rather than a duplicate of the LightGBM model.

CatBoost. CatBoost is added because its structure is different in two ways. First, it uses ordered boosting: training examples are processed through random permutations so that target-derived statistics for a row are computed only from earlier rows in that permutation [3]. This is important because `region_id`, month, and other calendar buckets are high-cardinality categorical signals that can otherwise memorize labels. Second, CatBoost commonly uses symmetric oblivious trees, where all nodes at the same depth share the same split. These balanced trees are less expressive per tree than arbitrary CART trees, but they are stable and provide a useful categorical-modeling bias for the ensemble.

The best reportable submission blends the three model families as

$$\hat{y}_{r,h} = 0.35\hat{y}_{r,h}^{LGB} + 0.35\hat{y}_{r,h}^{XGB} + 0.30\hat{y}_{r,h}^{CAT}. \tag{4}$$

All final predictions are clipped to $[0, 5]$.

D. Validation and Selection Discipline

We used three levels of evidence. First, chronological holdout and rolling-origin validation give fast local feedback. Second, Kaggle-like blind backtests hide a historical 91-day window and score the next five real weekly labels. Third, Kaggle public leaderboard readouts check transfer to the competition test distribution. These signals often disagreed, so no candidate was promoted from local MAE alone. We also used a leakage sentinel that poisons hidden-window labels and verifies that pseudo-test features do not change. This guard was important because a naive score-persistence model can look excellent locally but fail on Kaggle.

E. Inference and Reproducibility

At inference time, training tail rows and the 91-day test rows are concatenated per region, sorted by synthetic date, and transformed using the saved feature configuration. The last row of the test window becomes the forecast origin for each region. Every candidate submission is sanity-checked for 2,248 rows, exact sample-submission column order, matching `region_id` order, no NaN values, and prediction range $[0, 5]$. Each important Kaggle score in this report is tied to a CSV path, Kaggle reference id, blend description, and lineage label.

III. Experiments (2-4 pages)

A. Dataset and Setup

The dataset contains daily meteorological observations over many synthetic years for 2,248 regions. Training has 5,480

TABLE I
MAIN EXPERIMENT RESULTS AND DECISIONS

Experiment	Feature / model idea	Validation	Local MAE	Public MAE	Decision
Weather-only direct	Direct LGB/XGB without score history	Holdout	0.6770 / 0.7320	0.8640	Rejected as primary method; weather alone misses unobserved regional severity context.
LGB/XGB 50/50 anchor	Score-history boosted-tree ensemble	Holdout	0.6942 / 0.7150	0.8232	First reproducible legal anchor.
CatBoost tail2737	Direct CatBoost with native region/calendar categories	Rolling origin	0.2212	Component only	Useful diversity model; validation scale differs from holdout.
LGB/XGB/Cat 35/35/30	Three-model convex ensemble	Kaggle sanity + lineage audit	N/A	**0.8124**	**Best reportable method claim**, Kaggle ref 52698259.
Blind LGBM tail1095	Refit-full LGBM, lean features	Blind backtest	0.3549 blind	0.9380	Strong pseudo-private anchor but poor public transfer.
Feature-fused TCN	91-day sequence model with visible time-safe priors	Blind backtest	0.3508 blind	0.9450	Neural diversity signal only; not a public anchor.
Initial accidental hybrid	Leakage-adjacent score behavior	Diagnostic	0.2100	0.8094	Excluded from reportable claims.
Naive score persistence	score gap 0 / forward-fill control	Diagnostic	0.2321	1.0815	Demonstrates severe leakage risk.

days per region and weekly integer drought scores; test has 91 unlabeled days per region. Table II summarizes the data shape.

TABLE II
DATASET STATISTICS

Property	Train	Test
Days per region	5,480	91
Regions	2,248	2,248
Rows	12,319,040	204,568
Raw weather columns	14	14
Score range	0–5	N/A
Score NaN ratio	85.73%	N/A

Train-test distribution analysis showed that the real test window is not a simple continuation of the global training distribution. Compared with the recent 1,095-day training tail, test is hotter by about 5.35 degrees on average and slightly drier by precipitation and dry-day indicators. Against the stricter same-region, same-month-day previous-year baseline, daily precipitation drops from 3.250 to 1.925, dry-day share rises from 0.403 to 0.487, and 90-day precipitation falls from 277.676 to 223.710. This motivated tail-length, season-matching, and drought-proxy experiments rather than all-history training.

B. Main Model Results

Table I reports the major legal model families and diagnostic controls. Local MAE values come from different validation protocols, so they are used only within the same protocol. Kaggle public MAE is the common external comparison.

The key lesson is that local validation and public transfer are not monotonic. For example, rolling-origin CatBoost and later neural/TCN variants looked strong locally, but public leaderboard transfer was much worse than the historical LGB/XGB/CatBoost anchor. Conversely, the 35/35/30 ensemble made only a modest modeling change but improved the reproducible public score from 0.8232 to 0.8124.

C. Feature and Validation Ablations

Feature ablations with LightGBM lean tail1095 clarified which signals were essential. Removing score-history features degraded rolling MAE by about 0.052, confirming that old severity labels encode persistent regional drought state and unobserved geography. Removing long drought proxies caused a similar degradation, showing that accumulated rain deficit, dry streaks, and thermal stress are not redundant with short lags. Removing climatology improved rolling MAE by 0.0078, but failed the blind backtest with MAE 0.4463 versus the baseline blind anchor 0.3549. This was a concrete example where local validation alone would have selected the wrong model.

Missingness and shift experiments on 2026-05-23 also failed to replace the reportable anchor. The best validation/blind candidate, the season-matched LGBM shift run, reached rolling MAE 0.24574 and blind MAE 0.38370, but the recent pattern was that validation-strong families transferred poorly to public LB. We therefore kept it diagnostic rather than promoting it to the main claim.

D. Leaderboard and Final-Selection Readout

Table III shows the relevant public leaderboard snapshot after the final 2026-05-27 submissions. The public leaderboard is useful for Kaggle placement, but it can also reward overfitting to public feedback. Therefore we report two separate artifacts: the reportable model lineage and the selected public-chase artifact.

TABLE III
PUBLIC LEADERBOARD SNAPSHOT

Entry	Public MAE	Rank
Team 3	0.7628	1
Team 4	0.7817	2
Team 5 public-chase	**0.7915**	**3**
Team 20	0.7942	4
Team 2	0.7955	5
Team 1	0.8043	6
Baseline 3	0.8056	7
Team 5 reportable lineage	0.8124	–
Baseline 2	0.8623	–
Baseline 1	0.9117	–

The selected public-chase artifact is `submissions/baseline3_private_hedge_v3_cat35_lower_w1_more_horizon_0p225_0p375_0p55_0p75_1.csv` (Kaggle ref 53074980, SHA-12 003351676087, public MAE

0.7915). It is an exact-history horizon hedge moving the recovered 0.8094 public reference toward the clean 0.8124 anchor by horizon alphas $[0.225, 0.375, 0.55, 0.75, 1.00]$. It is useful for final selection, but it is not a new reportable training algorithm. The strongest same-day private-risk alternative is ref 53075022 at public MAE 0.7918, with additional cross-day alternatives at refs 53038040, 53038036, and 53038033.

The reportable artifact remains `submissions/ensemble_20260516_lgb_xgb_cat2737_35_35_30.csv` (Kaggle ref 52698259, SHA-12 bee6f618828d when restored locally). Its prediction distribution is valid and conservative: week means are approximately 0.955, 0.927, 0.880, 0.831, and 0.817, and all predictions are clipped within $[0, 5]$.

E. Discussion

The experiments reveal a "Kaggle paradox." Pure weather models and some validation-strong models look reasonable locally but underperform on the public leaderboard. Historical score features, when shifted safely beyond the 91-day blind gap, provide information about persistent regional vulnerability that raw weather alone cannot reconstruct. However, target history also creates the largest leakage risk, so the final workflow treats leakage sentinels, blind backtests, and exact artifact lineage as part of model selection rather than as after-the-fact documentation.

The public-chase sequence further shows why leaderboard scores must be labeled carefully. It improved Team 5 from the reportable 0.8124 anchor to 0.7915 by using public feedback around exact historical artifacts. This is acceptable for Kaggle final selection if the competition UI allows selecting the submitted artifact, but it should not be described as evidence that the supervised training method itself reached 0.7915.

IV. Conclusion

We built a leakage-aware drought severity forecasting pipeline for the Data Mining final project. The final reportable method uses direct-horizon LightGBM, XGBoost, and CatBoost models over carefully shifted temporal, meteorological, climatology, and historical severity features. The best reproducible modeling claim is the 35/35/30 boosted-tree ensemble with public MAE 0.8124. Extensive diagnostics showed that score history and long drought proxies are essential, that local rolling-origin MAE can be misleading under train-test shift, and that blind-window leakage controls are necessary. A separate public-chase artifact reaches public MAE 0.7915 and crosses Baseline 3; we keep it available for final leaderboard selection while preserving the clean 0.8124 lineage as the scientific method claim.

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